17 Given that

$$x^2 : (3x + 5) = 1 : 2$$

find the possible values of x.

$x^2 : (3x + 5) = 1 : 2$

	First	Second	
$\times x^2$	1	2	$\times x^2$
	x^2	$2x^2$	

1 mark
Form $2x^2 = 3x + 5$.

$2x^2 - 3x - 5 = 0$

1 mark
Rearrange to zero.

$2x^2 - 3x - 5 = (2x - 5)(x + 1)$

1 mark
Factorise the quadratic.

So $x = 5/2$ or $x = -1$.

1 mark
Both x values.

-1, 2.5

(Total for Question 17 is 4 marks)